Portfolio Optimization using Conditional Value-at-Risk

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Abstract

This project studies portfolio optimization for a portfolio of 10 NASDAQ stocks using Conditional Value-at-Risk and mean-CVaR optimization. We compare static, fixed-weight strategies with rolling monthly strategies, in which portfolio weights are updated each month based on recent historical data. The optimized portfolios are evaluated through back-testing and benchmarked against an equally weighted portfolio. The goal is to examine whether CVaR-based allocation methods can provide better downside-risk control and risk-adjusted performance than a simple passive baseline.

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1 Background & Preliminary

1.1 Risk Measures

Let Y denote the random variable representing portfolio loss, and let $\alpha \in (0,1)$ be the confidence level. The confidence level determines the tail region of the loss distribution: a higher α focuses on more extreme losses, while $1-\alpha$ represents the probability of losses exceeding the risk threshold.

Definition 1 (Value-at-Risk)**.** The Value-at-Risk at confidence level α, denoted by $\mathrm{VaR}_\alpha(Y)$, is the α-quantile of the loss distribution:

$$\mathrm{VaR}_\alpha(Y) = \inf\{\gamma \in \mathbb{R} : \mathbb{P}(Y \le \gamma) \ge \alpha\}.$$

Equivalently, for a continuous loss distribution,

$$\mathbb{P}(Y \ge \mathrm{VaR}_\alpha(Y)) = 1-\alpha.$$

Thus, $\mathrm{VaR}_\alpha(Y)$ represents the loss threshold that is exceeded only in the worst $1-\alpha$ proportion of cases.

Definition 2 (Conditional Value-at-Risk)**.** The Conditional Value-at-Risk at confidence level α, denoted by $\mathrm{CVaR}_\alpha(Y)$, is the expected loss conditional on losses exceeding the VaR threshold:

$$\mathrm{CVaR}_\alpha(Y) = \mathbb{E}[Y \mid Y \ge \mathrm{VaR}_\alpha(Y)]$$

While VaR identifies the beginning of the downside tail, CVaR measures the average severity of losses within that tail. Therefore, CVaR provides a more informative measure of extreme downside risk and is suitable for portfolio optimization.

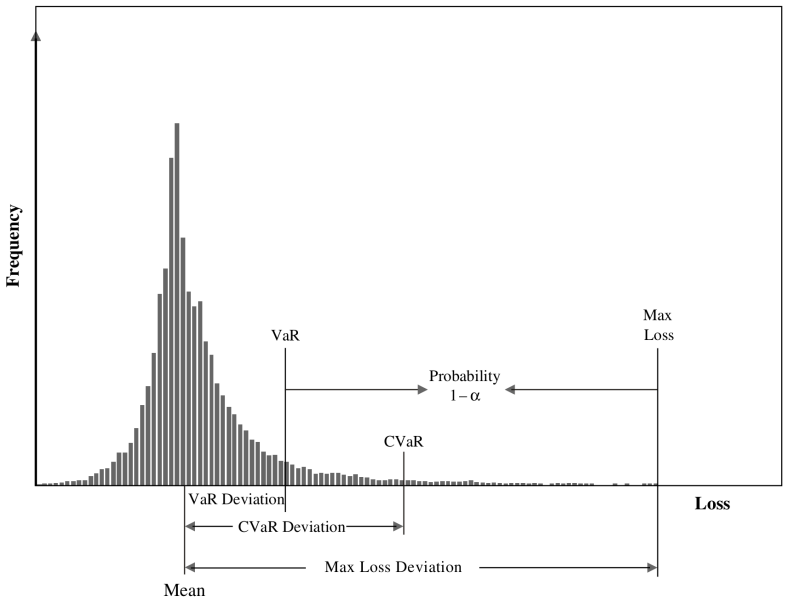


Figure 1: Graphical interpretation of VaR and CVaR for a portfolio loss distribution.

1.2 Linear Programming Formulation of CVaR and mean-CVaR

Proposition: CVaR Representation Let Y be a portfolio loss random variable and let $\alpha \in (0,1)$ be the confidence level. Then

$$\mathrm{CVaR}_\alpha(Y) = \min_\gamma \left[\gamma + \frac{1}{1-\alpha}\mathbb{E}[\max(Y-\gamma, 0)]\right].$$

The minimizer γ^* is the Value-at-Risk threshold:

$$\gamma^* = \mathrm{VaR}_\alpha(Y).$$

Here, γ is a candidate loss threshold, while $\max(Y-\gamma, 0)$ measures the excess loss beyond that threshold.

Proof. To construct the objective function of $\mathrm{CVaR}_\alpha(Y)$, proceed following steps:

Let Y be the random variable for the loss, and assume the distribution is continuous.

Denote $q = \mathrm{VaR}_\alpha(Y)$. By definition, we have:

$$\mathbb{P}(Y \le q) = \alpha \Rightarrow \mathbb{P}(Y > q) = 1 - \alpha$$

Since $\mathrm{CVaR}_\alpha(Y)$ is the average loss that exceeds q loss, that is:

$$\mathrm{CVaR}_\alpha(Y) = \mathbb{E}(Y|Y > q)$$

From the conditional expectation, we have:

$$\mathrm{CVaR}_\alpha(Y) = \mathbb{E}(Y \mid Y > q) = \frac{\mathbb{E}\left[Y\mathbf{1}_{\{Y>q\}}\right]}{\mathbb{P}(Y > q)}.$$

The indicator function of the event $\{Y > q\}$ is defined as

$$\mathbf{1}_{\{Y>q\}} = \begin{cases} 1, & \text{if } Y > q, \\ 0, & \text{if } Y \le q. \end{cases}$$

Hence, $Y\mathbf{1}_{\{Y>q\}}$ keeps only the losses that exceed the VaR threshold q, and sets other losses to 0. Since $q = \mathrm{VaR}_\alpha(Y)$, we have $\mathbb{P}(Y > q) = 1 - \alpha$. Hence,

$$\mathrm{CVaR}_\alpha(Y) = \frac{1}{1-\alpha}\mathbb{E}\left[Y\mathbf{1}_{\{Y>q\}}\right].$$

For the tail event $\{Y > q\}$, decompose the loss as

$$Y = q + (Y - q).$$

Therefore,

$$\begin{aligned} Y\mathbf{1}_{\{Y>q\}} &= q\mathbf{1}_{\{Y>q\}} + (Y-q)\mathbf{1}_{\{Y>q\}} \\ &= q\mathbf{1}_{\{Y>q\}} + \max(Y-q, 0). \end{aligned}$$

Taking the expectation on both sides gives:

$$\begin{aligned} \mathrm{CVaR}_\alpha(Y) &= \frac{1}{1-\alpha}\mathbb{E}\left[q\mathbf{1}_{\{Y>q\}} + \max(Y-q,0)\right] \\ &= \frac{1}{1-\alpha}\left[q\mathbb{P}(Y>q) + \mathbb{E}\left[\max(Y-q,0)\right]\right] \\ &= q + \frac{1}{1-\alpha}\mathbb{E}\left[\max(Y-q,0)\right]. \end{aligned}$$

Now define, for any candidate threshold γ,

$$F(\gamma) = \gamma + \frac{1}{1-\alpha}\mathbb{E}\left[\max(Y-\gamma,0)\right].$$

The function $F(\gamma)$ is convex because $\max(Y-\gamma, 0)$ is convex in γ. Moreover, for a continuous distribution,

$$F'(\gamma) = 1 - \frac{1}{1-\alpha}\mathbb{P}(Y > \gamma).$$

At $\gamma = q$, we have $\mathbb{P}(Y > q) = 1 - \alpha$, so

$$F'(q) = 1 - \frac{1-\alpha}{1-\alpha} = 0.$$

Since F is convex and $F'(q) = 0$, q minimizes F. Therefore,

$$q = \arg\min_{\gamma} \left\{ \gamma + \frac{1}{1-\alpha}\mathbb{E}\left[\max(Y-\gamma, 0)\right] \right\}.$$

Since $q = \mathrm{VaR}_\alpha(Y)$, we obtain

$$\mathrm{CVaR}_\alpha(Y) = \min_{\gamma} \left(\gamma + \frac{1}{1-\alpha}\mathbb{E}\left[\max(Y-\gamma, 0)\right] \right).$$

□

Proposition 1 (Theoretical Mean-CVaR)**.** Mean-CVaR optimization extends the CVaR minimization problem by incorporating the expected portfolio return.

Applying the CVaR representation to $Y(w)$, we obtain

$$\mathrm{CVaR}_\alpha(Y(w)) = \min_{\gamma} \left[\gamma + \frac{1}{1-\alpha}\mathbb{E}\left[\max(-R^T w - \gamma, 0)\right] \right].$$

The mean return constraint is

$$\mathbb{E}[R^T w] = \mu^T w \geq \rho,$$

Explanation: Let $R \in \mathbb{R}^n$ be the random return vector of the assets and let $w \in \mathbb{R}^n$ be the portfolio weight vector. The portfolio return is $R^T w$, while the corresponding portfolio loss is

$$Y(w) = -R^T w.$$

where $\mu = \mathbb{E}[R]$ and ρ is the target expected return.

Theoretical Stochastic Programming Framework

Let $R(\omega) \in \mathbb{R}^n$ be the random return vector and w be the portfolio weights. The portfolio loss is

$$Y(w, \omega) = -R(\omega)^T w.$$

The CVaR minimization problem is

$$\min_{w \in X} \text{CVaR}_\alpha(Y(w, \omega)).$$

Using the CVaR representation,

$$\text{CVaR}_\alpha(Y(w, \omega)) = \min_\gamma \left[\gamma + \frac{1}{1-\alpha} \mathbb{E}\left[\max(Y(w, \omega) - \gamma, 0)\right] \right].$$

Introduce $z(\omega)$ by

$$z(\omega) \geq Y(w, \omega) - \gamma, \qquad z(\omega) \geq 0.$$

Then

$$z(\omega) = \max(Y(w, \omega) - \gamma, 0)$$

at optimum. Thus, w and γ are first-stage decisions, while $z(\omega)$ is a second-stage recourse variable measuring excess loss after the market scenario is realized.

1.3 From Stochastic Programming to Linear Programming

In the theoretical CVaR framework, the portfolio loss is a random variable. Let $R \in \mathbb{R}^n$ denote the random return vector of the assets and let $w \in \mathbb{R}^n$ be the portfolio weight vector. The portfolio return is $R^T w$, so the corresponding portfolio loss is

$$Y(w) = -R^T w.$$

The CVaR representation is

$$\text{CVaR}_\alpha(Y(w)) = \min_\gamma \left[\gamma + \frac{1}{1-\alpha} \mathbb{E}\left[\max(Y(w) - \gamma, 0)\right] \right],$$

where γ is a candidate VaR threshold. At the optimum, γ becomes $\text{VaR}_\alpha(Y(w))$.

To remove the nonlinear positive-part function, introduce an auxiliary variable z such that

$$z \geq Y(w) - \gamma, \qquad z \geq 0.$$

Since z is minimized in the objective, the optimizer sets

$$z = \max(Y(w) - \gamma, 0)$$

at optimum. Hence, the stochastic CVaR problem can be written as a two-stage stochastic program:

$$\min_{w, \gamma} \left[\gamma + \mathbb{E}\big(Q(w, \gamma, R)\big) \right],$$

where

$$Q(w, \gamma, R) = \min_z \frac{1}{1-\alpha} z$$

subject to

$$z \geq -R^T w - \gamma, \qquad z \geq 0.$$

Here, w and γ are first-stage decisions, chosen before the market return is realized. The variable z is a second-stage recourse variable, which measures the excess loss after a return scenario is observed.

In practice, the unknown distribution of R is approximated using historical daily return scenarios r_t, $t = 1, \ldots, T$. Each trading day is treated as one empirical scenario with probability $1/T$. Therefore,

$$\mathbb{E}[z] \approx \frac{1}{T} \sum_{t=1}^{T} z_t.$$

Empirical CVaR Linear Program

Using historical return scenarios r_t, $t = 1, \ldots, T$, define portfolio loss by

$$y_t(w) = -r_t^T w.$$

The empirical CVaR optimization problem is

$$\min_{w,\gamma,z} \gamma + \frac{1}{(1-\alpha)T} \sum_{t=1}^{T} z_t$$

subject to

$$z_t \geq -r_t^T w - \gamma, \qquad z_t \geq 0, \qquad t = 1, \ldots, T,$$

$$\mathbf{1}^T w = 1, \qquad 0 \leq w_i \leq 1.$$

Empirical Mean-CVaR Linear Program

Mean-CVaR extends CVaR minimization by imposing a minimum expected return requirement. Let

$$\mu = \frac{1}{T} \sum_{t=1}^{T} r_t$$

be the sample mean return vector, and let ρ be the target expected return. The mean-CVaR problem is

$$\min_{w,\gamma,z} \gamma + \frac{1}{(1-\alpha)T} \sum_{t=1}^{T} z_t$$

subject to

$$\begin{aligned} & z_t \geq -r_t^T w - \gamma \\ & z_t \geq 0 \\ & \mu^T w \geq \rho \\ & \mathbf{1}^T w = 1, \\ & 0 \leq w_i \leq 1 \end{aligned}$$

2 Optimization Formulation

2.1 Linear Programming Formulation of CVaR

This section connects the theoretical CVaR formulation with the portfolio optimization problem in this project. Throughout the project, historical daily returns are treated as empirical market scenarios. The optimization model chooses one portfolio weight vector w that minimizes downside tail risk over the training window.

Symbol	Dimension / Type	Meaning
n	scalar	Number of stocks in the portfolio universe; here $n = 10$.
T	scalar	Number of historical trading days in the training window; each day is one scenario.
r_t	$\mathbb{R}^n$	Return vector of all stocks on day t, where $t = 1, \ldots, T$.
R_{train}	$\mathbb{R}^{T \times n}$	Training return matrix, with rows as trading days and columns as stocks.
w	$\mathbb{R}^n$	Portfolio weight vector; w_i is the allocation to stock i.
$y_t(w)$	scalar	Portfolio loss on day t, defined by $y_t(w) = -r_t^T w$.
α	scalar in $(0, 1)$	Confidence level for VaR/CVaR; here $\alpha = 0.95$.
γ	scalar	VaR threshold variable of the portfolio loss distribution.
z_t	scalar	Excess loss above the VaR threshold on day t.
μ	$\mathbb{R}^n$	Sample mean return vector from the training window.
ρ	scalar	Target expected return in the mean-CVaR model.
$\mathbf{1}$	$\mathbb{R}^n$	Vector of ones used for the full-investment constraint $\mathbf{1}^T w = 1$.

Table 1: Notation used in the CVaR and mean-CVaR portfolio formulation.

Empirical CVaR Optimization Model

Using historical return scenarios r_t, $t = 1, \ldots, T$, the portfolio loss is

$$y_t(w) = -r_t^T w.$$

The empirical CVaR optimization problem is:

$$\min_{w,\gamma,z} \gamma + \frac{1}{(1-\alpha)T} \sum_{t=1}^{T} z_t$$

subject to

$$\begin{aligned} z_t &\geq -r_t^T w - \gamma, \\ z_t &\geq 0, \\ \mathbf{1}^T w &= 1, \\ 0 \leq w_i &\leq 1 \end{aligned}$$

Empirical Mean-CVaR Optimization Model

Define the sample mean return vector:

$$\mu = \frac{1}{T}\sum_{t=1}^{T} r_t$$

And ρ be the target expected return. The empirical mean-CVaR problem is:

$$\min_{w,\gamma,z} \gamma + \frac{1}{(1-\alpha)T}\sum_{t=1}^{T} z_t$$

subject to

$$\begin{aligned} z_t &\geq -r_t^T w - \gamma, \\ z_t &\geq 0, \\ \mu^T w &\geq \rho, \\ \mathbf{1}^T w &= 1, \\ 0 \leq w_i &\leq 1 \end{aligned}$$

3 Methodology

3.1 Data & Return Matrix

Tickers: Consider a portfolio consisting of 10 NASDAQ-listed stocks:

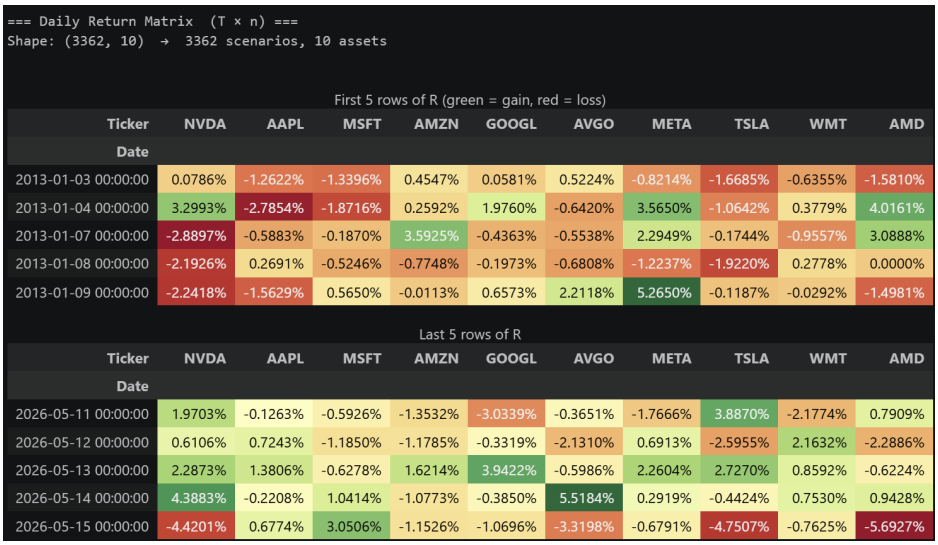
$$\{\text{NVDA}, \text{AAPL}, \text{MSFT}, \text{AMZN}, \text{GOOGL}, \text{AVGO}, \text{META}, \text{TSLA}, \text{WMT}, \text{AMD}\}.$$

Time Frame: Januray 1, 2013 - until now.

Returns Matrix Let $P_{t,i}$ denote the adjusted closing price of asset i on trading day t. The corresponding daily return is computed as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Use the formula above, compute the returns, we have the table as:



```
=== Daily Return Matrix  (T x n) ===
Shape: (3362, 10)  →  3362 scenarios, 10 assets
```

First 5 rows of R (green = gain, red = loss)

Ticker	NVDA	AAPL	MSFT	AMZN	GOOGL	AVGO	META	TSLA	WMT	AMD
Date										
2013-01-03 00:00:00	0.0786%	-1.2622%	-1.3396%	0.4547%	0.0581%	0.5224%	-0.8214%	-1.6685%	-0.6355%	-1.5810%
2013-01-04 00:00:00	3.2993%	-2.7854%	-1.8716%	0.2592%	1.9760%	-0.6420%	3.5650%	-1.0642%	0.3779%	4.0161%
2013-01-07 00:00:00	-2.8897%	-0.5883%	-0.1870%	3.5925%	-0.4363%	-0.5538%	2.2949%	-0.1744%	-0.9557%	3.0888%
2013-01-08 00:00:00	-2.1926%	0.2691%	-0.5246%	-0.7748%	-0.1973%	-0.6808%	-1.2237%	-1.9220%	0.2778%	0.0000%
2013-01-09 00:00:00	-2.2418%	-1.5629%	0.5650%	-0.0113%	0.6573%	2.2118%	5.2650%	-0.1187%	-0.0292%	-1.4981%

Last 5 rows of R

Ticker	NVDA	AAPL	MSFT	AMZN	GOOGL	AVGO	META	TSLA	WMT	AMD
Date										
2026-05-11 00:00:00	1.9703%	-0.1263%	-0.5926%	-1.3532%	-3.0339%	-0.3651%	-1.7666%	3.8870%	-2.1774%	0.7909%
2026-05-12 00:00:00	0.6106%	0.7243%	-1.1850%	-1.1785%	-0.3319%	-2.1310%	0.6913%	-2.5955%	2.1632%	-2.2886%
2026-05-13 00:00:00	2.2873%	1.3806%	-0.6278%	1.6214%	3.9422%	-0.5986%	2.2604%	2.7270%	0.8592%	-0.6224%
2026-05-14 00:00:00	4.3883%	-0.2208%	1.0414%	-1.0773%	-0.3850%	5.5184%	0.2919%	-0.4424%	0.7530%	0.9428%
2026-05-15 00:00:00	-4.4201%	0.6774%	3.0506%	-1.1526%	-1.0696%	-3.3198%	-0.6791%	-4.7507%	-0.7625%	-5.6927%

Figure 2: Daily return dataset of the selected NASDAQ stocks

For each trading day t, the vector of asset returns is written as

$$r_t = \begin{bmatrix} r_{t,1} & r_{t,2} & \cdots & r_{t,n} \end{bmatrix}^T,$$

where $n = 10$ is the number of stocks. Each daily return vector r_t is treated as one empirical scenario in the historical CVaR optimization framework.

The portfolio return on day t is

$$R_{p,t} = r_t^T w,$$

where

$$w = \begin{bmatrix} w_1 & w_2 & \cdots & w_n \end{bmatrix}^T$$

is the vector of portfolio allocation weights. Since CVaR is formulated in terms of losses, the portfolio loss on day t is defined as

$$y_t(w) = -r_t^T w.$$

Portfolio Return/Loss: Based on the above formula,consider a trading date, which is 15/5/2023, we can take this example of portfolio's return/loss on a historical trading (assume the weights are equal).

Thus, each historical trading day becomes one empirical scenario in the CVaR optimization problem. The model chooses one weight vector w that minimizes the tail losses of the portfolio across all training scenarios.

Scenario date: 2013-05-15

Ticker	daily_return	daily_return_pct
NVDA	0.0173	1.7301
AAPL	-0.0338	-3.3817
MSFT	0.0095	0.9544
AMZN	-0.0066	-0.6596
GOOGL	0.0325	3.2454
AVGO	0.0142	1.4243
META	-0.0174	-1.7362
TSLA	0.0192	1.9222
WMT	0.0137	1.3709
AMD	0.0282	2.8169

(a) Example returns of each stock

```
# Example: compute portfolio return/loss for one day using equal weights
example_weights = pd.Series(1 / len(TICKERS), index=TICKERS)

example_return = float(train_returns.iloc[scenario_id] @ example_weights)
example_loss = -example_return

print("Scenario date:", scenario_date.date())
print("Portfolio return:", f"{example_return:.4%}")
print("Portfolio loss:", f"{example_loss:.4%}")
```

0.0s

```
Scenario date: 2013-05-15
Portfolio return: 0.7687%
Portfolio loss: -0.7687%
```

(b) Portfolio Loss/Returns on that day

Figure 3: Example computation of portfolio's daily loss/returns

3.2 CVaR Linear Programming: Theory & Practice

3.2.1 Theoretical Computation

In this part, we discuss how we bridge those theoretical frameworks' parameters into our practice.

```
# — CELL: LP variable breakdown — show exactly what goes into the solver —
alpha = CONFIDENCE_LEVEL
T, n = train_returns.shape

print("=== LP Decision Vector  x = [w, γ, z₁…z_T] ===")
print(f"  Portfolio weights  w  : {n}  variables   (indices 0 … {n-1})")
print(f"  VaR threshold      γ  : 1   variable    (index  {n})")
print(f"  Excess losses      z  : {T}  variables   (indices {n+1} … {n+T})")
print(f"  ──────────────────────────────────────")
print(f"  Total variables      : {n + 1 + T}")
print(f"\nObjective coefficients c:")
print(f"  c[w]  = 0       (weights don't appear in objective directly)")
print(f"  c[γ]  = 1")
print(f"  c[z]  = 1 / ((1-α)·T) = {1/((1-alpha)*T):.6f}  (same for every z_t)")
print(f"\nNumber of inequality constraints (one per scenario): {T}")
print(f"  Form:  -r_t @ w - γ - z_t  ≤  0   for t = 1…{T}")
```

0.0s

```
=== LP Decision Vector  x = [w, γ, z₁…z_T] ===
  Portfolio weights  w  : 10  variables   (indices 0 … 9)
  VaR threshold      γ  : 1   variable    (index  10)
  Excess losses      z  : 2517  variables   (indices 11 … 2527)
  ──────────────────────────────────────
  Total variables      : 2528

Objective coefficients c:
  c[w]  = 0       (weights don't appear in objective directly)
  c[γ]  = 1
  c[z]  = 1 / ((1-α)·T) = 0.007946  (same for every z_t)

Number of inequality constraints (one per scenario): 2517
  Form:  -r_t @ w - γ - z_t  ≤  0   for t = 1…2517
```

(a) Codes to breakdown LP's components

(b) Output's of parameters

Figure 4: Linear Programming in practice

Figure 4a shows how the CVaR linear program is translated into the solver's input. For a training return matrix with $T = 2517$ trading days and $n = 10$ stocks, the decision vector is

$$x = \Big[w_1, \ldots, w_{10}, \gamma, z_1, \ldots, z_T\Big]^T.$$

Hence, the solver optimizes 10 portfolio weights, one VaR threshold γ, and 2517 excess-loss variables z_t. The objective function is

$$\min_{w,\gamma,z} \gamma + \frac{1}{(1-\alpha)T} \sum_{t=1}^{T} z_t.$$

Here, γ estimates the VaR threshold of the optimized portfolio, while each z_t measures the amount by which the portfolio loss on day t exceeds this threshold:

$$z_t = \max(-r_t^T w - \gamma, 0).$$

Although the weights w do not appear directly in the objective function, they determine the portfolio loss

on every training day through

$$y_t(w) = -r_t^T w.$$

Therefore, changing w changes the values of z_t, and hence changes the CVaR objective. In this sense, the solver minimizes CVaR by jointly choosing w, γ, and z_t.
For example, if on a certain day the portfolio loss is below the threshold,

$$-r_t^T w \leq \gamma,$$

then

$$z_t = 0.$$

If the loss exceeds the threshold,

$$-r_t^T w > \gamma,$$

then

$$z_t = -r_t^T w - \gamma.$$

Thus, z_t only becomes positive on tail-loss days. With 2517 training days, the model creates 2517 scenario constraints, one for each historical market scenario. The final optimized weights are the first 10 entries of the solver output.

3.2.2 Manual Computation

Consider the basic example with only 2 stocks, the intuition is:

Choosing $w \rightarrow$ different losses over each day $\rightarrow$ compute VaR $\rightarrow$ filter out by $z_t \rightarrow$ compute and choose minimal CVaR, corresponds to optimal w.

$$\alpha = 0.75, \qquad 1 - \alpha = 0.25.$$

Since there are $T = 4$ scenarios, the worst 25% corresponds to the worst one trading day.

Day	Stock A Return	Stock B Return
1	4%	1%
2	−10%	−3%
3	3%	1%
4	−8%	0%

Table 2: Toy historical return data.

Let w_A and w_B be the portfolio weights, where:

$$w_A + w_B = 1, \qquad 0 \leq w_A, w_B \leq 1.$$

For each candidate portfolio, the portfolio return on day t is computed as:

$$r_t^T w,$$

and the corresponding portfolio loss is

$$y_t(w) = -r_t^T w.$$

The empirical CVaR objective is

$$\gamma + \frac{1}{(1-\alpha)T} \sum_{t=1}^{T} z_t.$$

Since $(1-\alpha)T = 0.25(4) = 1$, the objective becomes

$$\gamma + \sum_{t=1}^{T} z_t.$$

For each day,

$$z_t = \max(y_t(w) - \gamma, 0),$$

so z_t is positive only when the portfolio loss exceeds the VaR threshold γ.
For simplicity, consider two stocks and four historical trading days. We set

$$\alpha = 0.75, \qquad 1-\alpha = 0.25, \qquad T = 4.$$

Hence,

$$(1-\alpha)T = 0.25(4) = 1.$$

Therefore, the empirical CVaR objective becomes

$$\gamma + \sum_{t=1}^{4} z_t.$$

All returns and losses below are written in percentage points.
Case 1: All capital allocated to Stock A

$$w_A = 1, \qquad w_B = 0.$$

The portfolio returns are

$$(4,\ -10,\ 3,\ -8).$$

Therefore, the portfolio losses are

$$(-4,\ 10,\ -3,\ 8).$$

Choose the VaR threshold

$$\gamma = 8.$$

The excess-loss variables are computed by

$$z_t = \max(y_t(w) - \gamma, 0).$$

Thus,

$$z = \big(\max(-4-8,0),\ \max(10-8,0),\ \max(-3-8,0),\ \max(8-8,0)\big).$$

So,

$$z = (0,\ 2,\ 0,\ 0).$$

The objective value is

$$\gamma + \sum_{t=1}^{4} z_t = 8 + (0+2+0+0) = 10.$$

Doing the same with following cases:

- **Case 2: Equally-weighted portfolio** $w_A = w_B = 0.5$.
- **Case 3: More capital allocated to Stock B** $w_A = 0.25, w_B = 0.75$.
- **Case 4: All capital allocated to Stock B** $w_A = 0, w_B = 1$.

Among the four tested candidates, the smallest objective value is obtained by allocating fully to Stock B:

$$w_A = 0, \qquad w_B = 1.$$

This portfolio has the lowest empirical CVaR because it produces the smallest worst-case historical loss.

Candidate	w_A	w_B	Losses $y_t(w)$	γ	Objective
All A	1.00	0.00	$(-4,\ 10,\ -3,\ 8)$	8	$8 + 2 = 10$
Equal Weight	0.50	0.50	$(-2.5,\ 6.5,\ -2,\ 4)$	4	$4 + 2.5 = 6.5$
More B	0.25	0.75	$(-1.75,\ 4.75,\ -1.5,\ 2)$	2	$2 + 2.75 = 4.75$
All B	0.00	1.00	$(-1,\ 3,\ -1,\ 0)$	0	$0 + 3 = 3$

Table 3: Manual comparison of candidate portfolios under the empirical CVaR objective.

This example illustrates the role of each decision variable. The weights w determine the portfolio losses across historical scenarios. The threshold γ identifies where the tail-loss region begins. The variables z_t measure the amount by which each day's loss exceeds γ. Therefore, the optimizer does not minimize weights directly; it chooses weights that make

$$\gamma + \frac{1}{(1-\alpha)T} \sum_{t=1}^{T} z_t$$

as small as possible.

3.2.3 Simplex Method Computation

The following table gives a *Simplex-style* interpretation. The Simplex Method does not randomly test portfolios; it moves from one feasible vertex of the LP region to another vertex that improves the objective. For this toy example, we demonstrate this idea by moving across several candidate portfolios.

Step	w_A	w_B	Losses $y_t(w)$	γ	Objective
1	1.00	0.00	$(-4,\ 10,\ -3,\ 8)$	8	$8 + 2 = 10$
2	0.50	0.50	$(-2.5,\ 6.5,\ -2,\ 4)$	4	$4 + 2.5 = 6.5$
3	0.25	0.75	$(-1.75,\ 4.75,\ -1.5,\ 2)$	2	$2 + 2.75 = 4.75$
4	0.00	1.00	$(-1,\ 3,\ -1,\ 0)$	0	$0 + 3 = 3$

Table 4: Simplex-style movement toward portfolios with lower empirical CVaR.

For each candidate, the solver conceptually checks how the weights affect the portfolio losses. Then it chooses the VaR threshold γ, computes the excess-loss variables z_t, and evaluates

$$\gamma + \sum_{t=1}^{4} z_t.$$

For example, in Step 3:

$$w_A = 0.25, \qquad w_B = 0.75.$$

The portfolio losses are

$$(-1.75,\ 4.75,\ -1.5,\ 2).$$

Choose

$$\gamma = 2.$$

Then

$$z = \big(\max(-1.75 - 2, 0),\ \max(4.75 - 2, 0),\ \max(-1.5 - 2, 0),\ \max(2 - 2, 0)\big).$$

Therefore,

$$z = (0,\ 2.75,\ 0,\ 0).$$

The objective value is

$$2 + (0 + 2.75 + 0 + 0) = 4.75.$$

Moving from Step 1 to Step 4 gives

$$10 \to 6.5 \to 4.75 \to 3.$$

Thus, the best candidate in this simplified illustration is

$$w_A^* = 0, \qquad w_B^* = 1.$$

In the actual Simplex Method, these portfolios are not manually selected. The algorithm uses the LP constraint matrix to move between feasible vertices and stops when no neighboring vertex improves the objective.

3.2.4 Interior Point Method Computation

The Interior Point Method follows a different idea. Instead of moving from vertex to vertex, it starts from an interior feasible point and moves through the inside of the feasible region toward the optimum. The following table is a simplified numerical illustration of this direction of movement.

Step	w_A	w_B	Losses $y_t(w)$	γ	Objective
1	0.50	0.50	$(-2.5,\ 6.5,\ -2,\ 4)$	4.00	6.50
2	0.25	0.75	$(-1.75,\ 4.75,\ -1.5,\ 2)$	2.00	4.75
3	0.10	0.90	$(-1.3,\ 3.7,\ -1.2,\ 0.8)$	0.80	3.70
4	0.02	0.98	$(-1.06,\ 3.14,\ -1.04,\ 0.16)$	0.16	3.14
5	0.00	1.00	$(-1,\ 3,\ -1,\ 0)$	0.00	3.00

Table 5: Interior-point-style movement toward the optimal CVaR solution.

For example, at Step 3:

$$w_A = 0.10, \qquad w_B = 0.90.$$

The portfolio returns are

$$(1.3,\ -3.7,\ 1.2,\ -0.8).$$

Therefore, the portfolio losses are

$$(-1.3,\ 3.7,\ -1.2,\ 0.8).$$

Choose

$$\gamma = 0.8.$$

Then

$$z = \big(\max(-1.3 - 0.8, 0),\ \max(3.7 - 0.8, 0),\ \max(-1.2 - 0.8, 0),\ \max(0.8 - 0.8, 0)\big).$$

So,

$$z = (0,\ 2.9,\ 0,\ 0).$$

The objective value is

$$0.8 + (0 + 2.9 + 0 + 0) = 3.7.$$

As the method moves closer to $w_A = 0$, the objective decreases:

$$6.5 \to 4.75 \to 3.70 \to 3.14 \to 3.00.$$

Hence, the solution approached by the Interior Point Method is also

$$w_A^* = 0, \qquad w_B^* = 1,$$

with

$$\gamma^* = 0, \qquad z^* = (0,\ 3,\ 0,\ 0),$$

and objective value

$$3.$$

3.2.5 Remark on those Computations

The computations above are only a simplified demonstration of how CVaR minimization behaves. The four manually chosen portfolios are not all possible portfolios. Since portfolio weights are continuous, there are infinitely many feasible choices of w_A and w_B satisfying

$$w_A + w_B = 1, \qquad 0 \le w_A, w_B \le 1.$$

In the actual project, the model has 10 stocks and thousands of historical trading-day scenarios. The LP solver does not manually test a small list of candidate weights. Instead, it solves for

$$w, \gamma, z_1, \ldots, z_T$$

simultaneously using the full system of linear constraints. The Simplex Method can be understood as moving across vertices of the feasible region, while the Interior Point Method moves through the interior of the feasible region. Both methods aim to find the same optimal solution:

$$(w^*, \gamma^*, z^*).$$

3.3 Solver Output

After solving the CVaR linear program, the optimal decision vector is

$$x^* = \left[w_1^*, \ldots, w_n^*, \gamma^*, z_1^*, \ldots, z_T^*\right]^T.$$

The first n entries give the optimized portfolio weights, γ^* gives the VaR threshold, and z_t^* gives the excess loss above this threshold for each training day.

Figure 5 shows that the optimized portfolio has a 95% VaR threshold of approximately 1.6875% daily loss and a CVaR value of approximately 2.6014%. Among 2517 training scenarios, 127 days are identified as tail-loss days, matching the worst 5% of the sample. The excess-loss variable is computed as

$$z_t^* = \max(y_t(w^*) - \gamma^*, 0),$$

so z_t^* is positive only when the portfolio loss exceeds the VaR threshold. This confirms how the LP captures tail-risk severity in practice.

```python
# — CELL: Show what γ* and z* look like after solving ————————————
weights, info = solve_cvar_lp(train_returns, alpha=CONFIDENCE_LEVEL)

scenario_losses = -train_returns.to_numpy() @ weights.to_numpy()   # y_t = -r_t^T w*
gamma_star      = np.quantile(scenario_losses, CONFIDENCE_LEVEL)    # VaR = γ*
z_star          = np.maximum(scenario_losses - gamma_star, 0)       # z_t* = max(y_t - γ*, 0)
cvar_check      = gamma_star + z_star.mean() / (1 - CONFIDENCE_LEVEL)

tail_mask = scenario_losses >= gamma_star

print(f"=== LP Solution Decomposition ===")
print(f"  γ*  (VaR at {int(CONFIDENCE_LEVEL*100)}%)      = {gamma_star:.4%}  daily loss threshold")
print(f"  CVaR (objective value)  = {cvar_check:.4%}  average loss in worst {int((1-CONFIDENCE_LEVEL)*100)}% of days")
print(f"  Tail days (z_t > 0)     : {tail_mask.sum()} of {len(scenario_losses)} scenarios ({tail_mask.mean():.1%})")
print(f"  Mean z_t across tail    : {z_star[z_star>0].mean():.4%}")

# Show the worst 8 scenarios
worst_idx = np.argsort(scenario_losses)[::-1][:8]
worst_df = pd.DataFrame({
    "Date": train_returns.index[worst_idx],
    "Portfolio loss  y_t": scenario_losses[worst_idx],
    "Excess above γ*  z_t": z_star[worst_idx],
    "In CVaR tail?": tail_mask[worst_idx],
}).set_index("Date")
display(
    worst_df.style
    .format({"Portfolio loss  y_t": "{:.3%}", "Excess above γ*  z_t": "{:.3%}"})
    .set_caption("8 worst in-sample scenarios — these define CVaR")
)
```

(a) LP solution decomposition code

```
=== LP Solution Decomposition ===
  γ*  (VaR at 95%)      = 1.6875%  daily loss threshold
  CVaR (objective value)  = 2.6014%  average loss in worst 5% of days
  Tail days (z_t > 0)     : 127 of 2517 scenarios (5.0%)
  Mean z_t across tail    : 0.9201%
```

8 worst in-sample scenarios — these define CVaR

Date	Portfolio loss y_t	Excess above γ* z_t	In CVaR tail?
2020-03-12 00:00:00	9.082%	7.394%	True
2020-03-16 00:00:00	8.738%	7.051%	True
2022-05-17 00:00:00	7.026%	5.339%	True
2018-02-20 00:00:00	6.805%	5.117%	True
2015-10-14 00:00:00	6.750%	5.063%	True
2022-05-18 00:00:00	6.166%	4.479%	True
2022-07-26 00:00:00	5.935%	4.247%	True
2020-03-11 00:00:00	4.682%	2.995%	True

(b) Optimized CVaR tail-loss output

Figure 5: Decomposition of the optimized CVaR solution into VaR threshold, CVaR value, and tail-loss scenarios.

3.4 Backtesting Design

After obtaining the optimized portfolio weights from the training period, we evaluate each strategy on an unseen test period. This out-of-sample evaluation is called backtesting. It allows us to examine whether the optimized allocation remains effective on data that were not used during the optimization process.

We compare five strategies: Equal Weight, Static CVaR, Rolling Monthly CVaR, Static Mean-CVaR, and Rolling Monthly Mean-CVaR. The equal-weight portfolio serves as the benchmark, with

$$w_i = \frac{1}{10}, \qquad i = 1, \ldots, 10.$$

In the static strategies, the optimization problem is solved once using the full training period. The resulting weights are then held fixed throughout the entire test period. In the rolling monthly strategies, the optimization problem is re-solved at the beginning of each test month using the most recent historical return window. The newly optimized weights are applied to the following month.

For each strategy, daily out-of-sample portfolio returns are computed by

$$R_{p,t} = r_t^T w.$$

Cumulative wealth is then calculated as

$$W_t = \prod_{s=1}^{t} (1 + R_{p,s}).$$

The strategies are evaluated using cumulative wealth, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, VaR, and CVaR. These metrics allow us to compare both return performance and downside-risk control across the different allocation methods.

4 Results and Discussion

4.1 Out-of-Sample Performance Summary

Out-of-Sample Performance Summary

The table below evaluates realized performance over the 3-year out-of-sample period only.

```
metrics = performance_metrics(strategy_returns, alpha=CONFIDENCE_LEVEL, risk_free_rate=RISK_FREE_RATE)
metrics = metrics.loc[strategy_order]
display(format_metrics(metrics))
```

strategy	cumulative_return	annual_return	annual_volatility	sharpe_ratio	max_drawdown	daily_var_95_loss	daily_cvar_95_loss
Equal Weight	2.725807	0.552978	0.283019	1.95	-0.300192	0.026615	0.039346
CVaR 95% Static	2.026782	0.448658	0.180153	2.49	-0.234072	0.015608	0.023482
CVaR 95% Rolling Monthly	1.821885	0.415071	0.173066	2.40	-0.217468	0.015340	0.022810
Mean-CVaR 95% Static	4.262455	0.743227	0.331222	2.24	-0.320854	0.028260	0.045964
Mean-CVaR 95% Rolling Monthly	3.964349	0.709536	0.245631	2.89	-0.257653	0.022356	0.031774

Figure 6: Strategy Performance Summary

The optimized portfolios are evaluated on the unseen three-year out-of-sample period. Table **??** summarizes the realized performance of the equal-weight benchmark, pure CVaR strategies, and mean-CVaR strategies.

Overall, mean-CVaR strategies achieve the strongest cumulative growth. The static mean-CVaR strategy obtains the highest cumulative return, approximately 426%, while rolling monthly mean-CVaR achieves approximately 396%. Both outperform the equal-weight benchmark, which achieves approximately 273%. However, the higher return of static mean-CVaR comes with the highest annualized volatility, approximately 33.1%, and the largest maximum drawdown, approximately −32.1%.

Pure CVaR strategies are more conservative. Static CVaR and rolling monthly CVaR produce lower cumulative returns, but they also have lower volatility, lower VaR, lower CVaR, and smaller maximum drawdowns than the equal-weight benchmark. This confirms that pure CVaR optimization is effective for downside-risk control, but may sacrifice return growth.

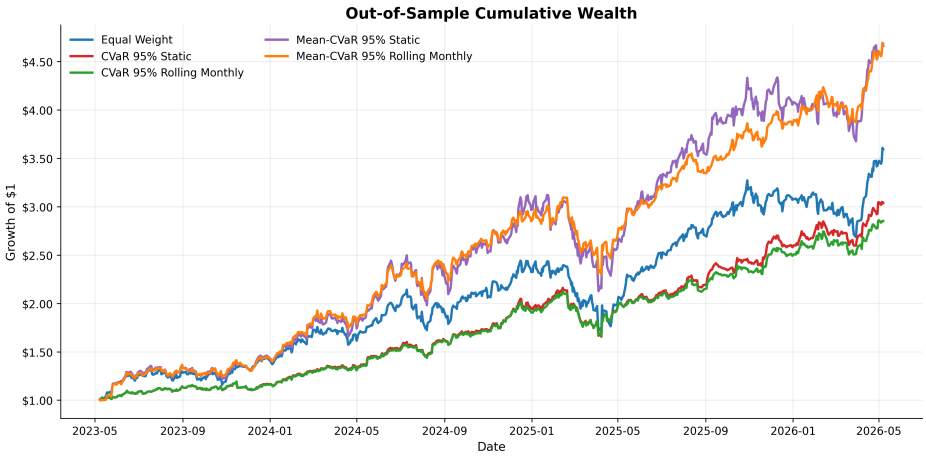


Figure 7: Out-of-sample cumulative wealth across equal-weight, CVaR, and mean-CVaR strategies.

Figure 7 shows that mean-CVaR strategies dominate in cumulative wealth, while pure CVaR strategies remain more defensive. This result suggests that adding an expected-return constraint prevents the optimizer from becoming overly conservative.

4.2 Portfolio Allocation Behavior

Figure 8 compares the initial allocation of the equal-weight, static CVaR, and static mean-CVaR portfolios. The equal-weight portfolio assigns 10% to each stock, while the optimized portfolios allocate capital unevenly according to the estimated downside-risk and return structure.

The pure CVaR portfolio allocates heavily toward defensive assets, especially WMT, because its objective only minimizes downside tail risk. In contrast, the mean-CVaR portfolio includes more exposure to growth-oriented stocks such as NVDA and AVGO, since the expected-return constraint requires the portfolio to maintain a minimum return level.

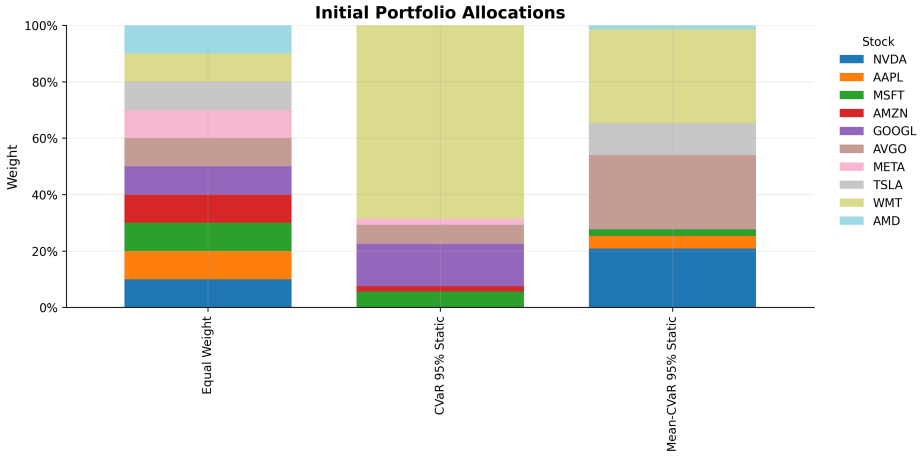
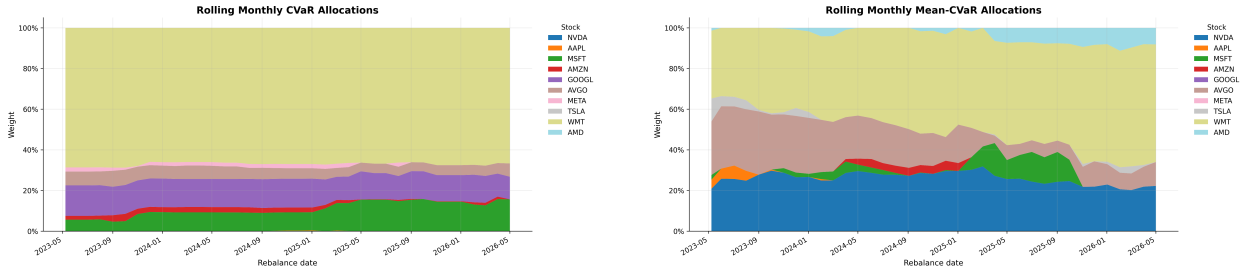


Figure 8: Initial portfolio allocations for equal-weight, static CVaR, and static mean-CVaR strategies.

The rolling allocation plots further show how the optimized weights change through time. Rolling CVaR remains relatively stable and defensive, with large exposure to WMT. Rolling mean-CVaR is more dynamic, allocating more capital to higher-growth stocks while still maintaining downside-risk awareness.



(a) Rolling Monthly CVaR

(b) Rolling Monthly Mean-CVaR

Figure 9: Rolling monthly portfolio allocations under CVaR and mean-CVaR strategies.

4.3 Risk-Return and Downside-Risk Analysis

Figure 10 compares the annualized return and annualized volatility of each strategy. Static mean-CVaR achieves the highest annualized return, approximately 74.3%, but also has the highest volatility, approximately 33.1%. This makes it the most aggressive portfolio.

Rolling monthly mean-CVaR provides the most practical risk-return tradeoff. It achieves a strong annualized return of approximately 71.0% with lower volatility, approximately 24.6%, and the highest Sharpe ratio, approximately 2.89. Therefore, rolling mean-CVaR performs better as a balanced strategy between growth and risk control.

Pure CVaR strategies have lower annualized returns, approximately 44.9% for static CVaR and 41.5% for

rolling CVaR. However, they also have the lowest volatility and lower downside-risk measures. Their daily CVaR losses are approximately 2.35% and 2.28%, compared with 3.93% for the equal-weight benchmark.

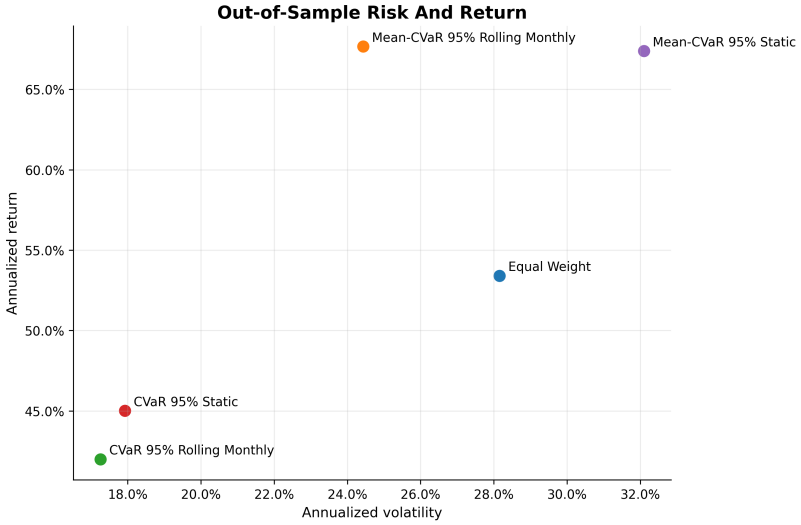


Figure 10: Out-of-sample annualized return versus annualized volatility.

The drawdown plot confirms the same tradeoff. Mean-CVaR strategies achieve stronger growth but experience deeper drawdowns during stress periods. Pure CVaR strategies sacrifice return but provide more stable downside-risk protection.

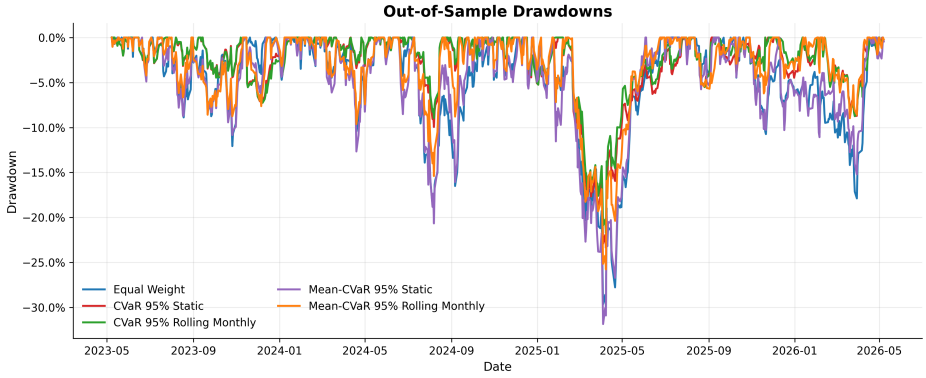


Figure 11: Out-of-sample drawdowns across portfolio strategies.

4.4 Interpretation and Limitations

The training-period mean-CVaR efficient frontier illustrates the central tradeoff in this project: higher target expected return requires accepting higher historical CVaR loss. This explains why mean-CVaR strategies outperform in growth but also carry higher volatility and larger drawdowns.

The main result is that pure CVaR optimization constructs defensive portfolios with lower volatility and lower tail-risk measures, while mean-CVaR optimization improves return performance by adding a minimum expected-return requirement. Among all tested strategies, rolling monthly mean-CVaR gives the most attractive practical balance because it achieves high return, lower volatility than static mean-CVaR, and the highest Sharpe ratio.

However, the project has several limitations. First, the optimization relies on historical returns, so the results depend on the assumption that past scenarios are informative for future risk. Second, the portfolio is long-only and restricted to ten selected NASDAQ stocks, which limits diversification. Third, transaction

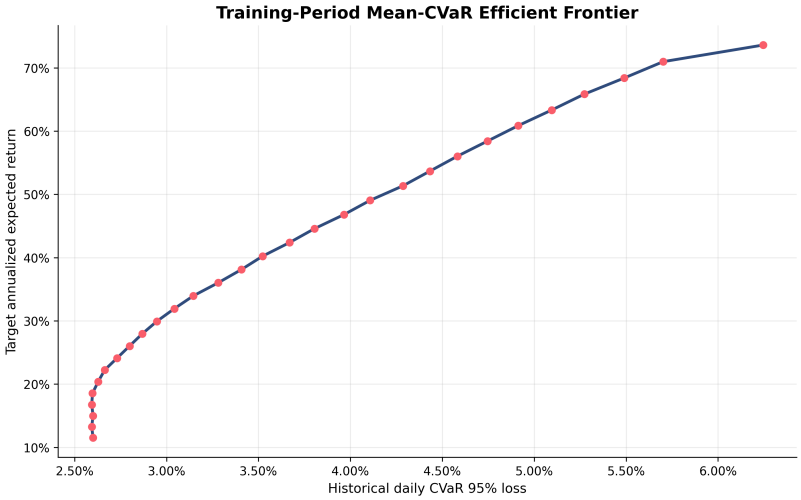


Figure 12: Training-period mean-CVaR efficient frontier.

costs, taxes, and market impact are not included in the backtest. Finally, rolling monthly rebalancing may improve adaptiveness, but in practice it would also introduce trading costs.

5 Conclusion

5.1 Summary of Findings

This project formulates CVaR and mean-CVaR portfolio allocation as linear programming problems using historical daily returns as empirical scenarios. The model optimizes portfolio weights, the VaR threshold, and scenario-level excess losses to control downside tail risk.

The results show that pure CVaR strategies are more defensive: they reduce volatility and tail-risk measures, but generate lower cumulative returns. Mean-CVaR strategies achieve stronger out-of-sample growth because the minimum expected-return constraint prevents the optimizer from becoming overly conservative. Rolling monthly mean-CVaR provides the best practical tradeoff, combining strong return performance with lower risk than the static mean-CVaR strategy.

5.2 Future Extensions

The current project is based on a long-only buy-and-hold stock portfolio. Future extensions can allow short selling, include transaction costs and taxes, or expand the asset universe to bonds, ETFs, options, and other derivatives. More advanced frameworks could also incorporate hedging, leverage constraints, robust optimization, or multi-period portfolio optimization. These extensions would make the model more realistic, but would require deeper portfolio theory and risk-management techniques.

References

[1] Rockafellar, R. T., and Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. *Journal of Risk*.

[2] Cornuejols, G., and Tütüncü, R. *Optimization Methods in Finance*.